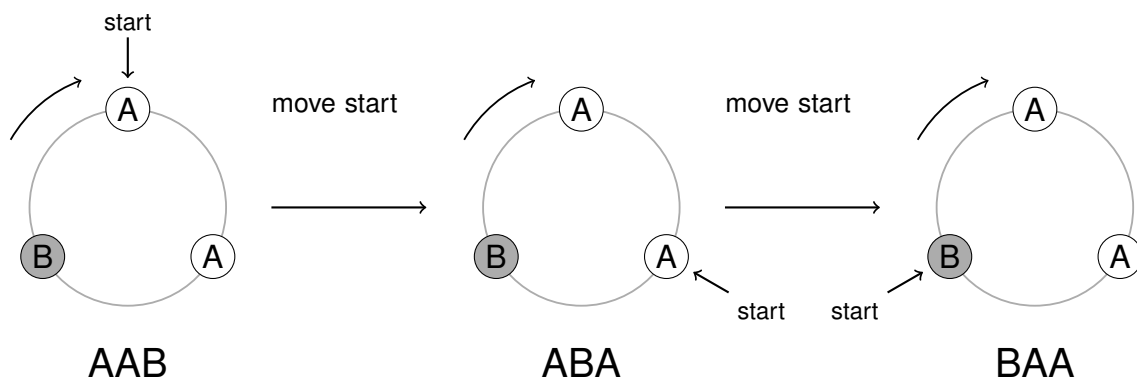
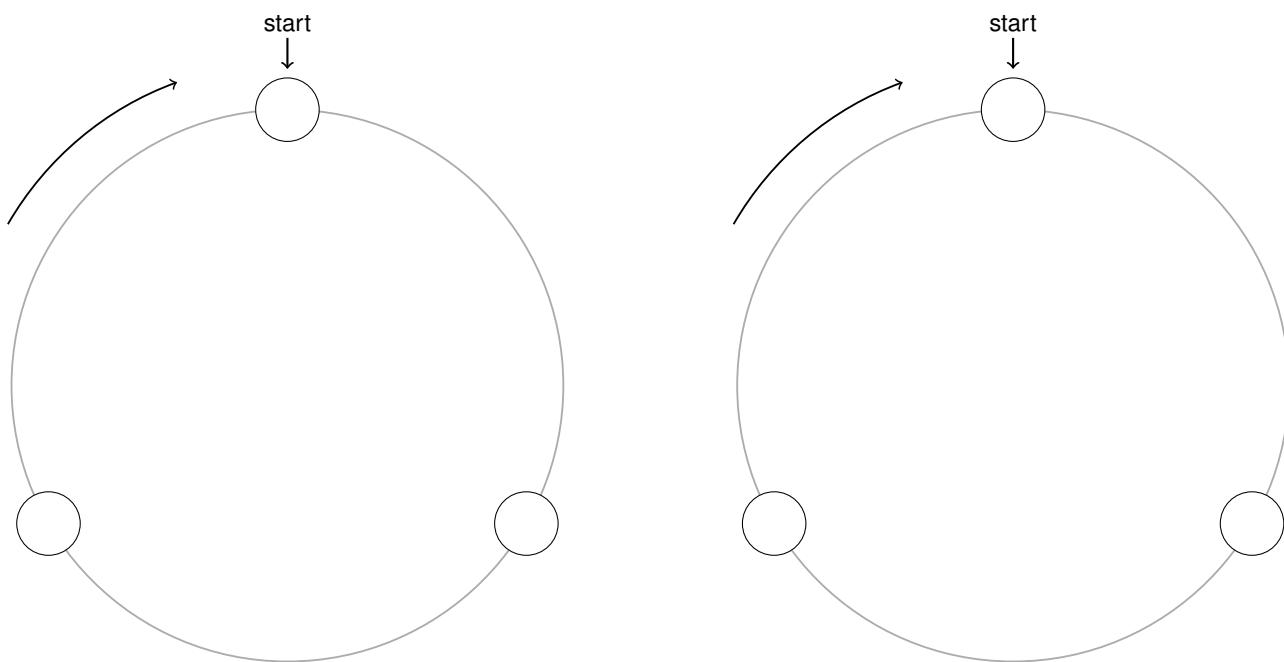


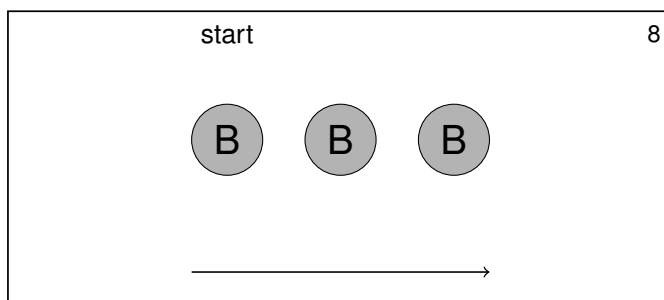
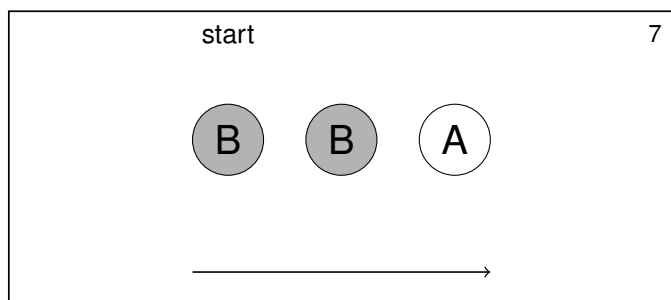
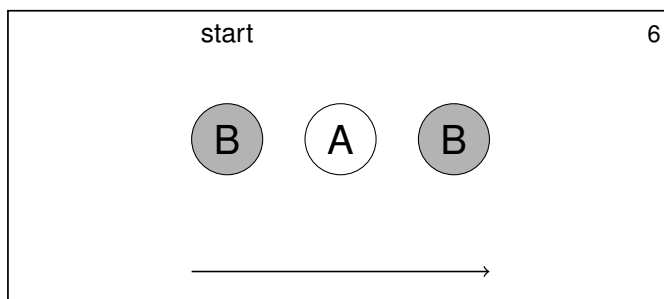
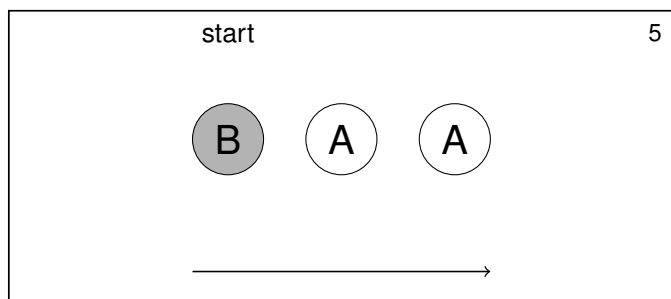
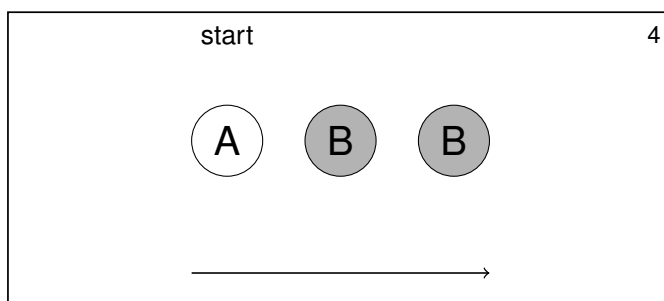
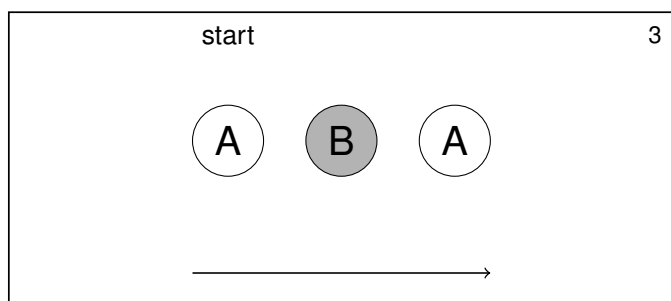
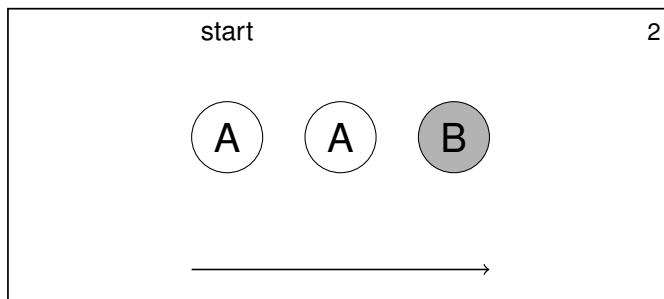
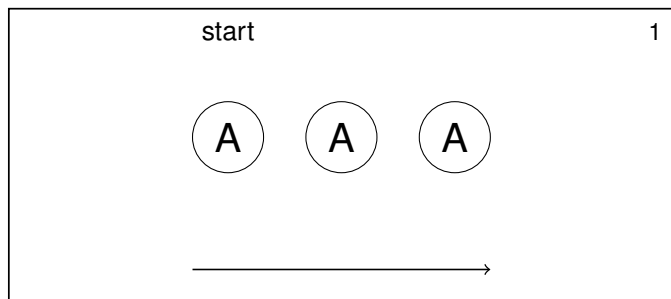
Each bead may be A or B; one-color rings are allowed. Read clockwise from the start arrow. The bead order is the readout. Rings count as the same when turning makes them match. Keep them face up; do not flip them over.



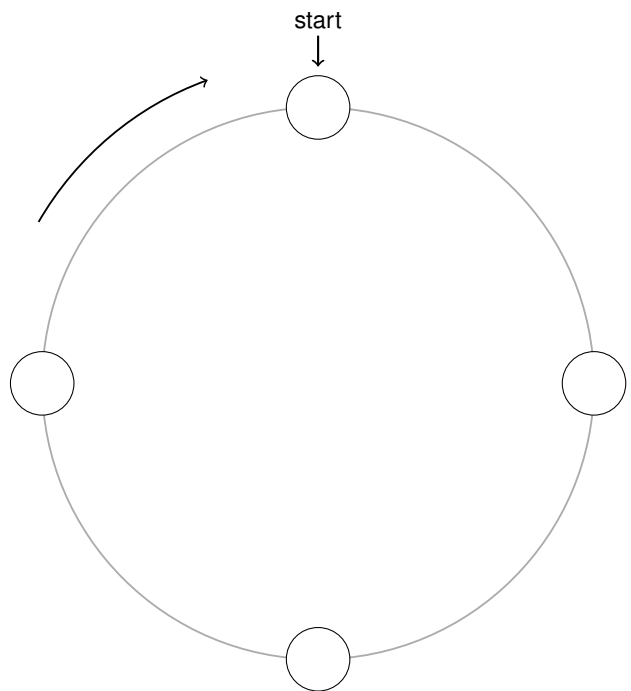
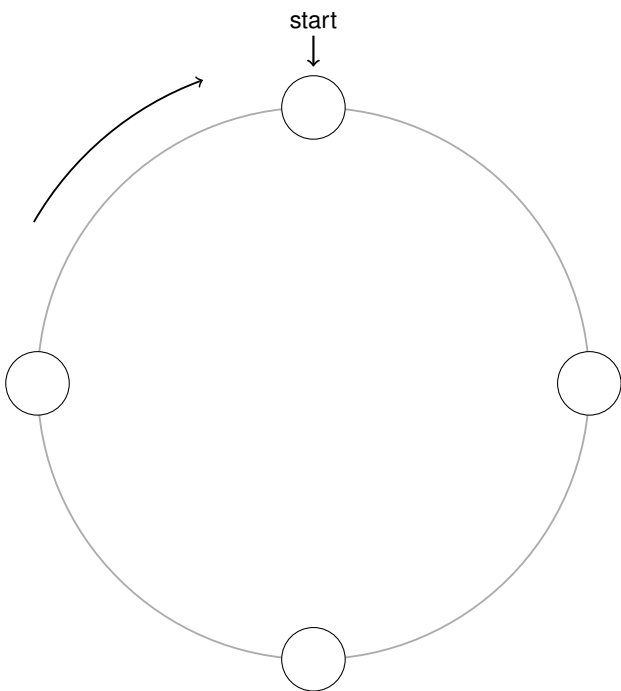
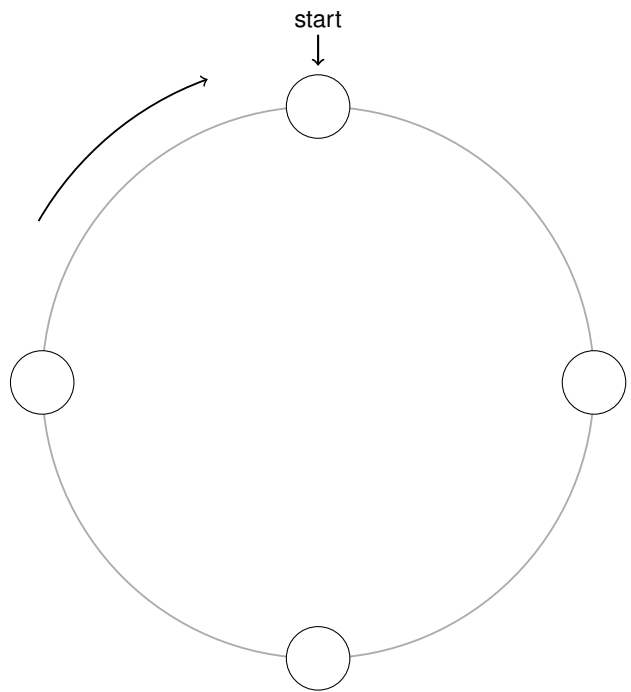
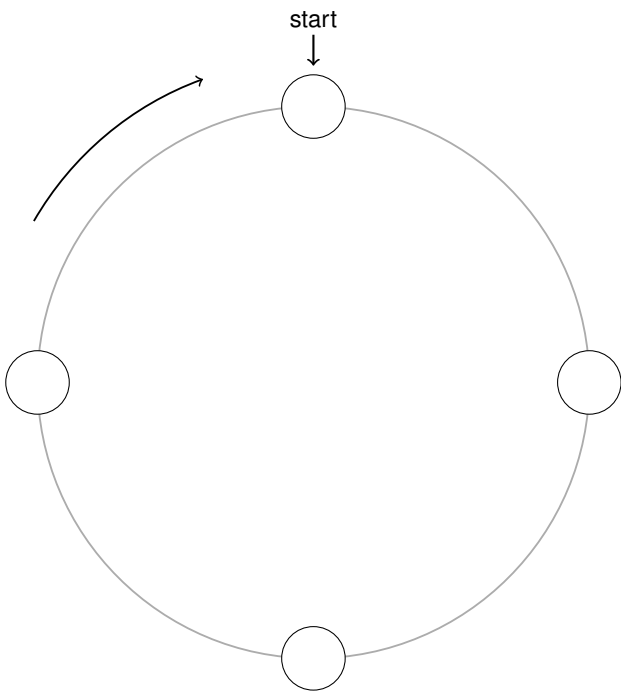
Problem 1: Make every different three-bead ring using A and B. Find all its readouts as you move the start.



Problem 2: Group cards that make the same ring. How many groups are there, and how many cards are in each group?



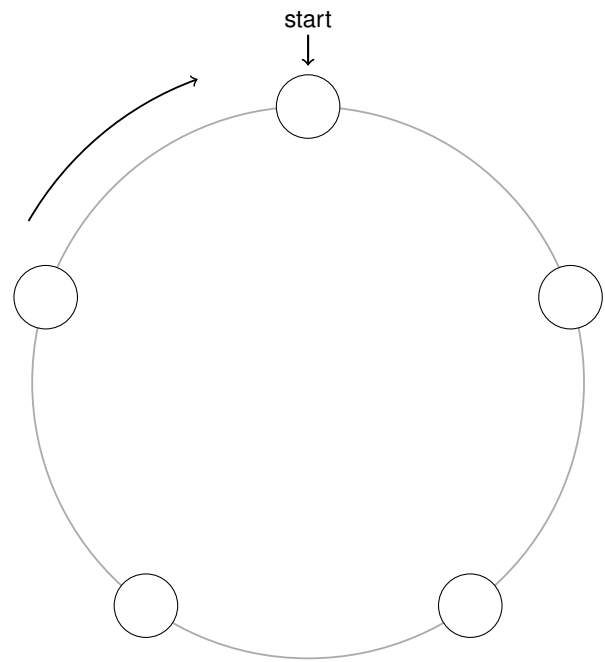
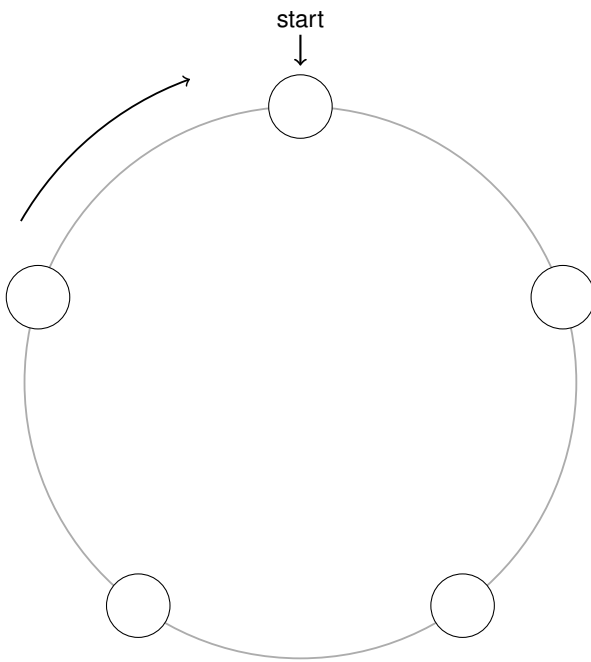
Problem 3: Make four-bead rings with exactly one, two, three or four different readouts. For each impossible number, explain why it cannot happen.



Problem 4: Group these five-bead cards by the rings they make. Find the size of each group and the total number of different rings.

start 1 	start 2 	start 3 	start 4
start 5 	start 6 	start 7 	start 8
start 9 	start 10 	start 11 	start 12
start 13 	start 14 	start 15 	start 16
start 17 	start 18 	start 19 	start 20
start 21 	start 22 	start 23 	start 24
start 25 	start 26 	start 27 	start 28
start 29 	start 30 	start 31 	start 32

Problem 5: Can a five-bead ring using both A and B have fewer than five different readouts? Find one or explain why it cannot happen.



Problem 6: Someone divides 32 by 5 to count the five-bead rings. What went wrong? Find a calculation that counts them correctly.
